

ECON 5345 Homework 5 Report

Harlly Zhou

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1. The following table reports the results of the lag selection. The AIC chooses $p = 6$, the BIC chooses $p = 1$, and the likelihood ratio test chooses $p = 3$ (We reject $p = 4$ at the 5% significance level).

Table 1: Lag selection statistics for the VAR model

p	AIC	BIC	LR p -value
1	-1.456476	-1.026065	0.0157
2	-1.445090	-0.670350	0.0000
3	-1.609263	-0.490194	0.0005
4	-1.680275	-0.216877	0.0536
5	-1.636079	0.171648	0.0004
6	-1.710671	0.441384	0.1895
7	-1.627121	0.869263	0.0036
8	-1.651718	1.188995	—

2. The following table reports the least squares estimates of the $VAR(6)$ model. Each cell reports the coefficient with significance stars and the t -statistic in parentheses.

Table 2: $VAR(6)$ least squares estimates.

RHS regressor	$\Delta \log(\text{Real GDP})$	$\Delta \log(\text{GDP Deflator})$	$\Delta \log(\text{EXR})$	FFR
Constant	0.2803 (1.145)	-0.0043 (-0.059)	-0.7676 (-0.738)	-0.6635* (-2.221)
$\Delta \log(\text{Real GDP})_{t-1}$	0.3590*** (3.496)	0.0218 (0.705)	-0.1342 (-0.307)	0.4713*** (3.761)
$\Delta \log(\text{GDP Deflator})_{t-1}$	-0.5061. (-1.669)	0.5533*** (6.055)	-1.0817 (-0.840)	-0.2412 (-0.652)
$\Delta \log(\text{EXR})_{t-1}$	0.0027 (0.114)	-0.0134. (-1.895)	0.3885*** (3.895)	-0.0347 (-1.211)
FFR_{t-1}	0.1120 (1.420)	0.0766** (3.223)	-0.0502 (-0.150)	1.0867*** (11.291)
$\Delta \log(\text{Real GDP})_{t-2}$	0.1610 (1.467)	-0.0690* (-2.086)	-0.5039 (-1.080)	0.2789* (2.082)
$\Delta \log(\text{GDP Deflator})_{t-2}$	1.4582*** (4.169)	0.1695 (1.608)	1.4276 (0.960)	1.3122** (3.074)
$\Delta \log(\text{EXR})_{t-2}$	-0.0191 (-0.759)	-0.0049 (-0.640)	-0.1293 (-1.207)	-0.0221 (-0.718)
FFR_{t-2}	-0.4387*** (-3.861)	-0.0796* (-2.323)	0.0062 (0.013)	-0.4988*** (-3.597)
$\Delta \log(\text{Real GDP})_{t-3}$	-0.0752 (-0.649)	0.0352 (1.006)	1.1853* (2.406)	0.1018 (0.719)
$\Delta \log(\text{GDP Deflator})_{t-3}$	-0.9872** (-2.707)	-0.0423 (-0.384)	-1.4989 (-0.967)	-0.6605 (-1.484)
$\Delta \log(\text{EXR})_{t-3}$	0.0308 (1.252)	-0.0044 (-0.595)	0.1781. (1.706)	-0.0139 (-0.463)
FFR_{t-3}	0.2735* (2.289)	0.0273 (0.759)	-0.0169 (-0.033)	0.5663*** (3.883)
$\Delta \log(\text{Real GDP})_{t-4}$	0.1806. (1.711)	0.0868** (2.726)	-0.0549 (-0.122)	-0.1657 (-1.286)
$\Delta \log(\text{GDP Deflator})_{t-4}$	0.0639 (0.171)	0.3038** (2.698)	2.6168 (1.648)	-0.4141 (-0.908)
$\Delta \log(\text{EXR})_{t-4}$	-0.0097 (-0.400)	-0.0011 (-0.151)	-0.0770 (-0.749)	0.0106 (0.360)
FFR_{t-4}	-0.0054 (-0.045)	-0.0628. (-1.727)	-0.5041 (-0.983)	-0.2130 (-1.446)
$\Delta \log(\text{Real GDP})_{t-5}$	0.0896 (0.890)	0.0179 (0.589)	-0.5004 (-1.170)	0.1938 (1.578)
$\Delta \log(\text{GDP Deflator})_{t-5}$	0.6285 (1.601)	-0.1428 (-1.207)	-3.3585* (-2.013)	0.0800 (0.167)
$\Delta \log(\text{EXR})_{t-5}$	-0.0030 (-0.126)	-0.0073 (-1.004)	-0.0830 (-0.813)	-0.0472 (-1.611)
FFR_{t-5}	-0.0774 (-0.649)	0.0781* (2.174)	1.5854** (3.130)	0.3486* (2.396)
$\Delta \log(\text{Real GDP})_{t-6}$	-0.1052 (-1.131)	-0.0343 (-1.224)	0.1793 (0.454)	-0.0700 (-0.616)
$\Delta \log(\text{GDP Deflator})_{t-6}$	-0.5657. (-1.779)	0.1020 (1.064)	2.1000 (1.554)	0.3177 (0.819)
$\Delta \log(\text{EXR})_{t-6}$	0.0232 (0.984)	0.0013 (0.179)	-0.0010 (-0.010)	0.0113 (0.392)
FFR_{t-6}	0.1217 (1.447)	-0.0421. (-1.662)	-0.9697** (-2.713)	-0.3526*** (-3.436)

Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, . $p < 0.1$. t -statistics in parentheses.

3. The following figure reports the IRF together with the error bands with parametric MC and bootstrap.

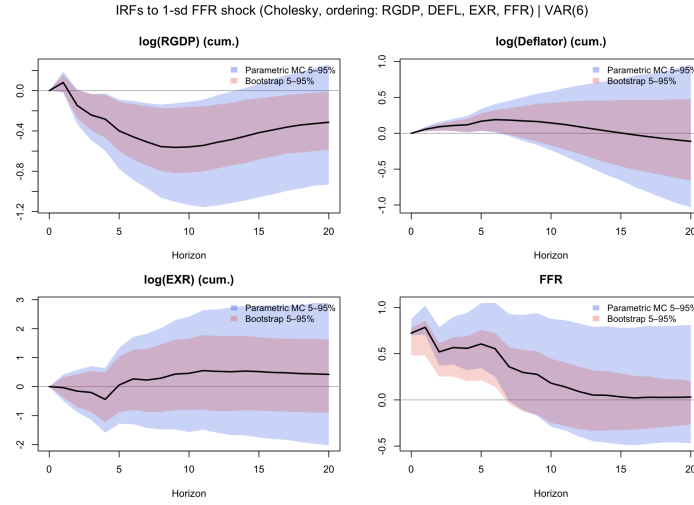


Figure 1: IRF for the $VAR(6)$ model with error bands with parametric MC and bootstrap.

The price level increases in response to the contractionary monetary policy shock. This is called the price puzzle. The $VAR(6)$ model only contains a limited number of variables, which does not exhaust all the factor affecting price level. There is also price stickiness and the cost channel which work through the opportunity cost of holding inventories.

The FEVD is plotted in Figure 2.

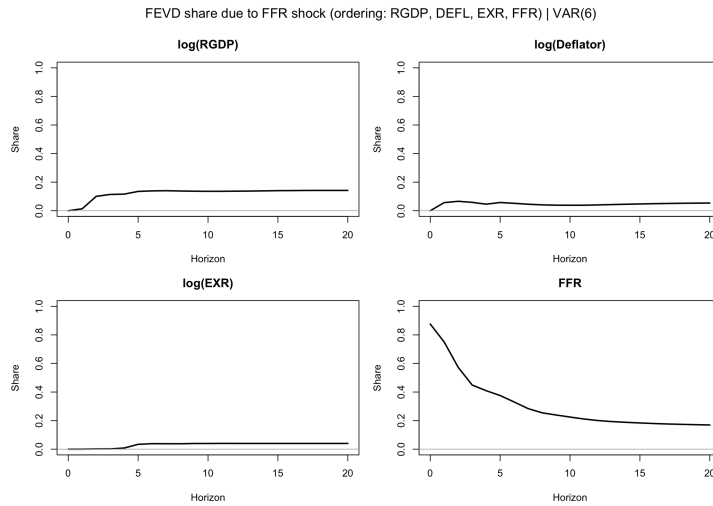


Figure 2: FEVD for the $VAR(6)$ model.

4. The new order results are plotted in Figure 3.

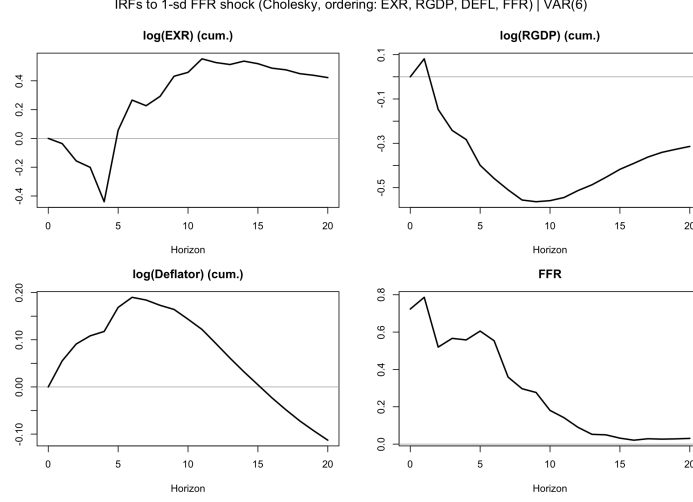


Figure 3: IRF for the $VAR(6)$ model with new order.

The IRFs are similar to the previous results. According to the paper, the Cholesky ordering tends to matter for where the policy instrument is. In these two questions, the policy instrument FFR is always put as the last variable, and the ordering only changes for the remaining variables. Therefore, the result does not change a lot.

5. The empirical responses contradicts with the Bornbusch model and the Eichenbaum-Evans paper. The EXR index at time t , I_t , is given by

$$I_t = I_{t-1} \prod_{j=1}^{J_t} \left(\frac{e_{j,t}}{e_{j,t-1}} \right)^{\omega_{j,t}},$$

where J_t is the number of foreign currencies, $e_{j,t}$ is the price of US dollars **in terms of the foreign currency j** , and $\omega_{j,t}$ is the weight of the foreign currency j in the basket.¹ So when the exchange rate increases, the US dollar appreciates. The overshooting result argues that when FFR increases, the US dollar should appreciate and keep appreciating for a while. However, the empirical responses in question 3 and 4 shows that the US dollar depreciates for a year and then starts to appreciate back.

Two possible reasons are as follows

- i. In their paper, commodity prices are included, while we do not. Commodity prices are forward-looking variables, which may capture some of the Fed's information when they consider the FFR as an instrument.

¹See <https://www.federalreserve.gov/pubs/feds/2000/200006/200006pap.pdf> for this definition.

- ii. The paper was in the 1990s. In particular, the Eichenbaum-Evans papers used the data from 1974 to 1990. Therefore, I also plotted the IRF for this period. And We do observe an appreciation after 1 quarter. However, there is still a considerable size of depreciation after around 1 year.

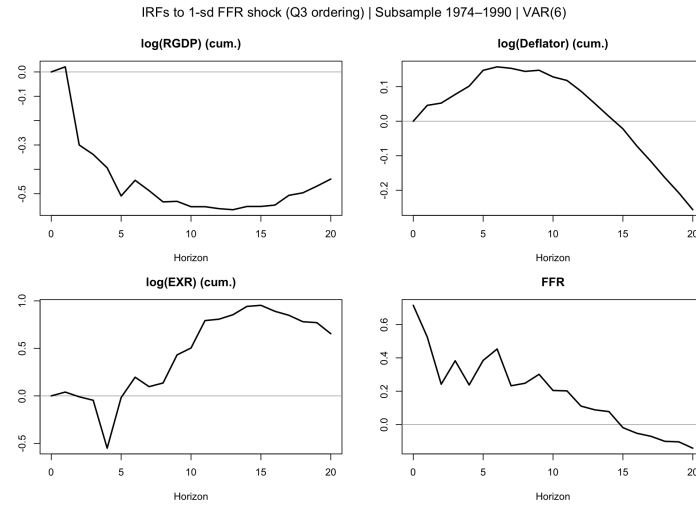


Figure 4: IRF for 1974-1990 data.